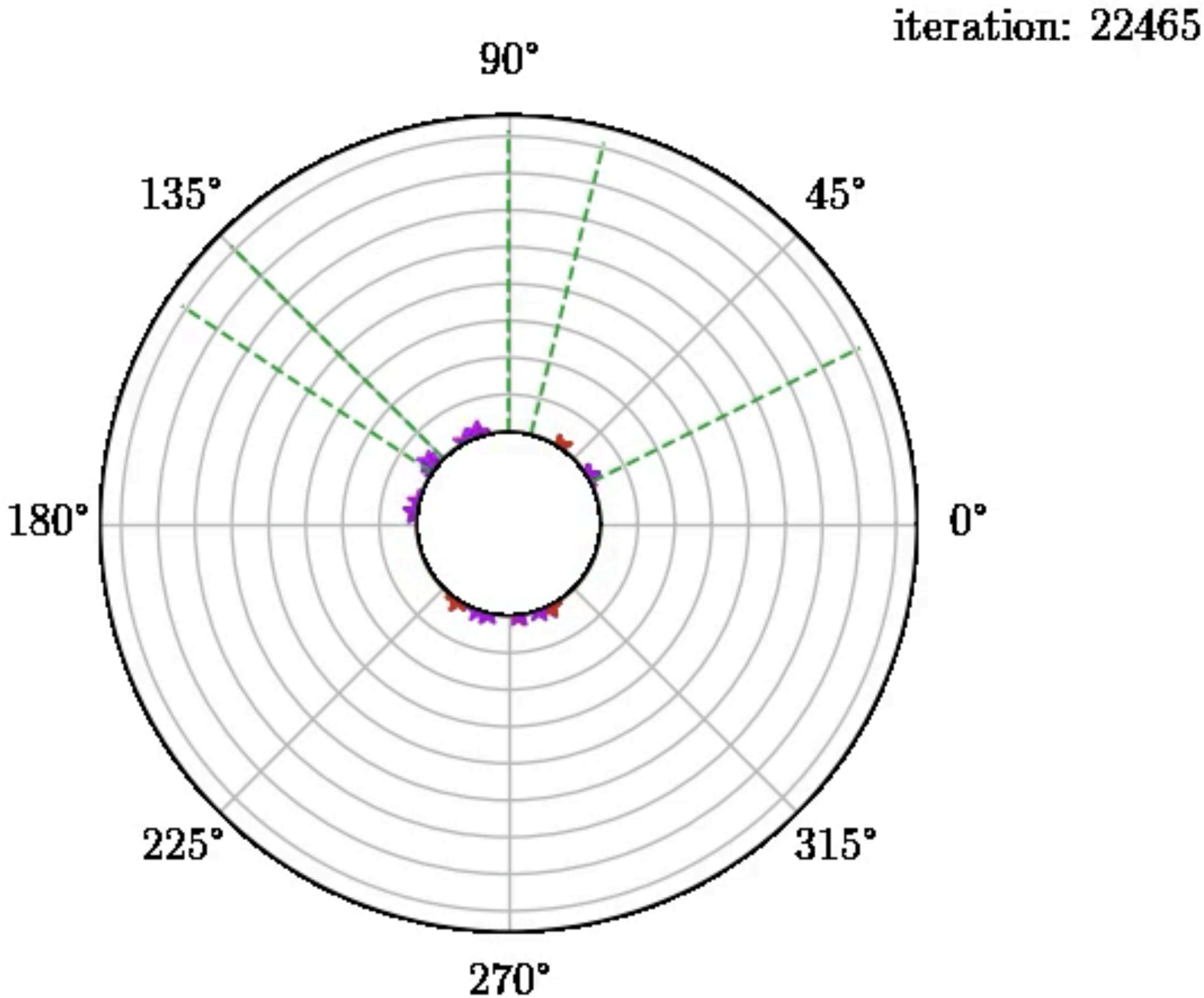
Illustration

20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90°
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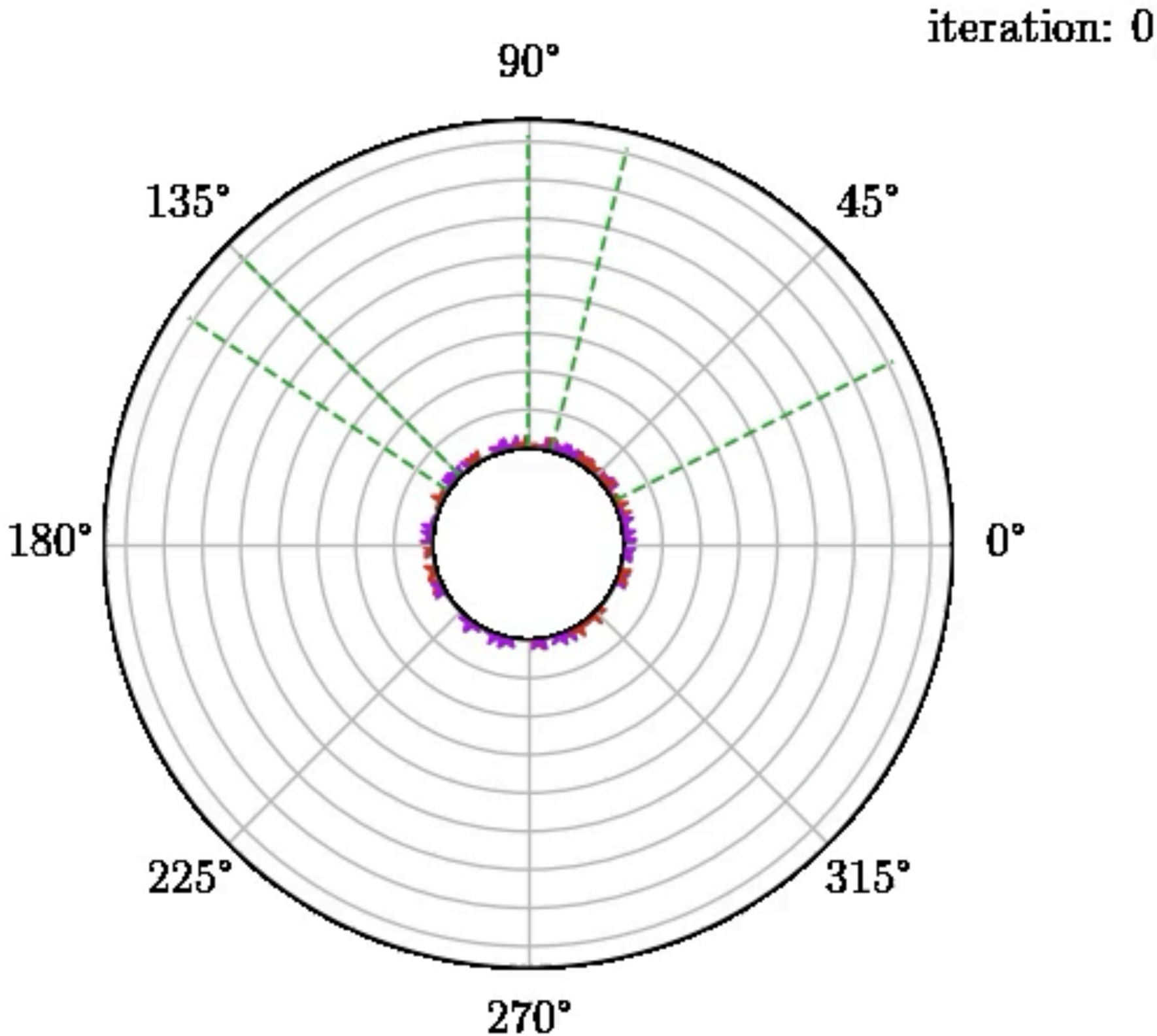
45°
180°
0°
225°
315°
270°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45°
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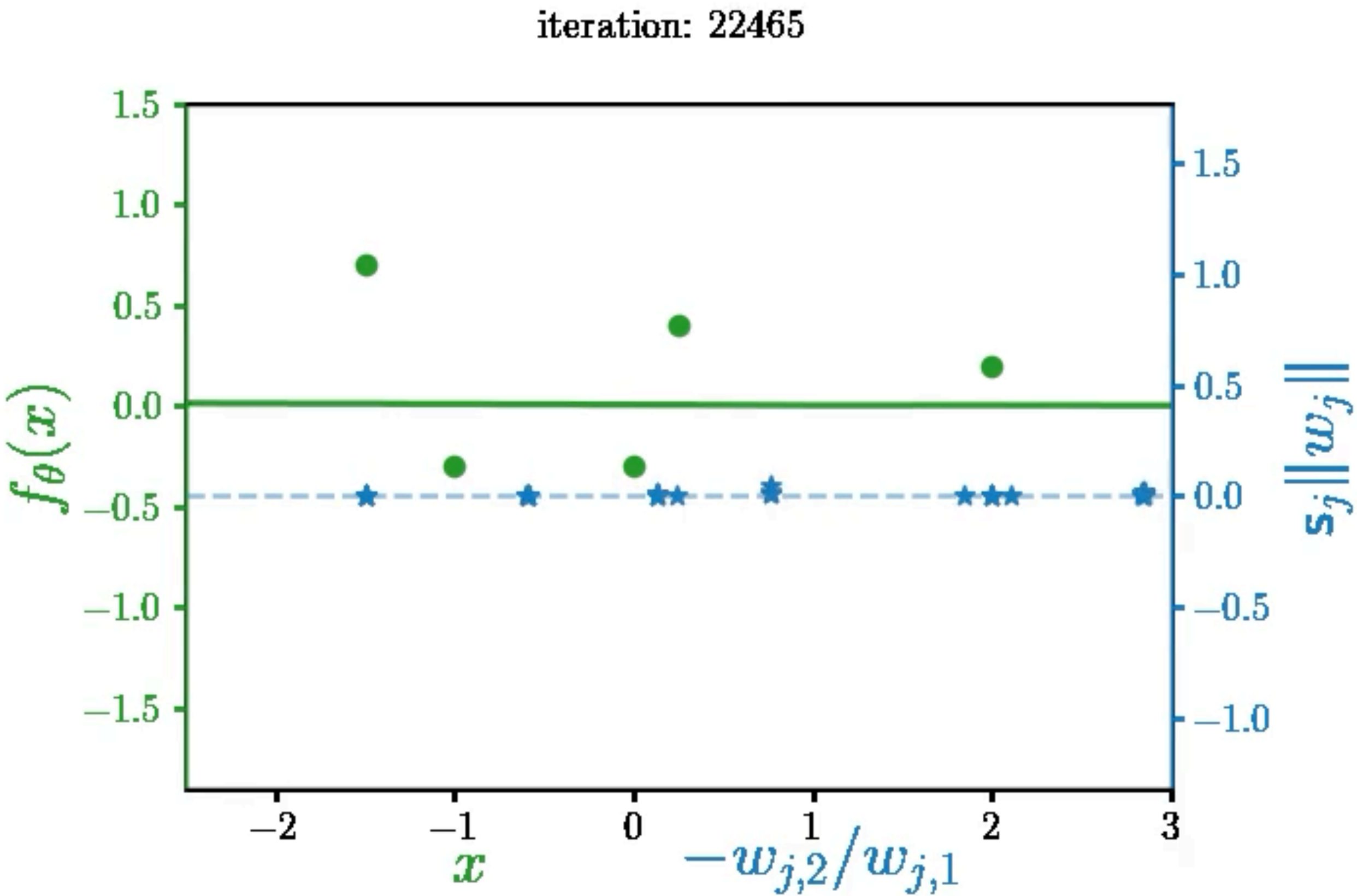
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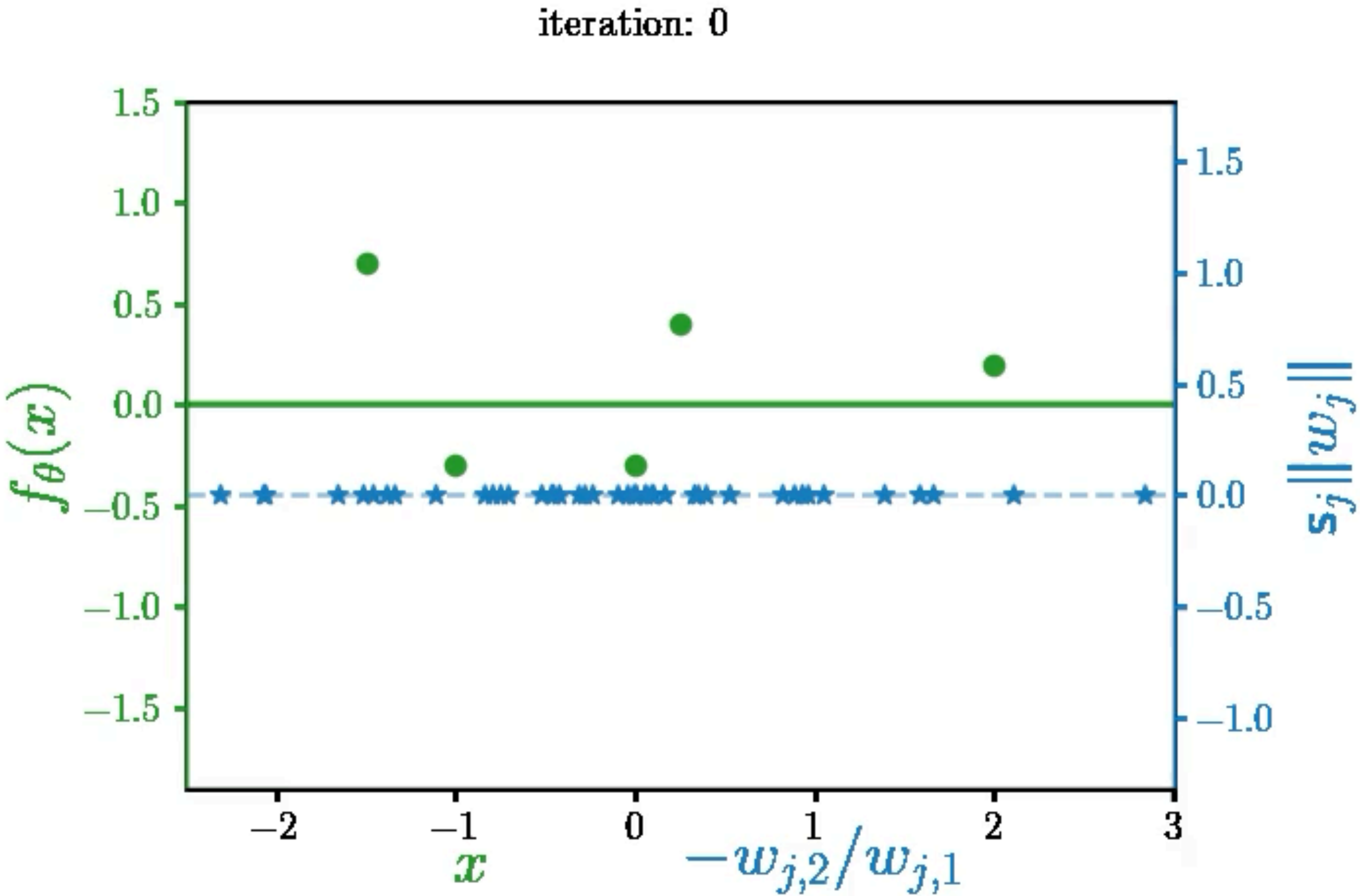
0.5
0.0
−0.5
−1.0
−1.5
$f_\theta(x)$
1.5
1.0
0.5
0.0
−0.5
−1.0
$s_j\|w_j\|$
−2
−1
0
1
2
3
x
$-w_{j,2}/w_{j,1}$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0.5
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−1.5
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1.5
1.0
0.5
0.0
−0.5
−1.0
$s_j\|w_j\|$
−2
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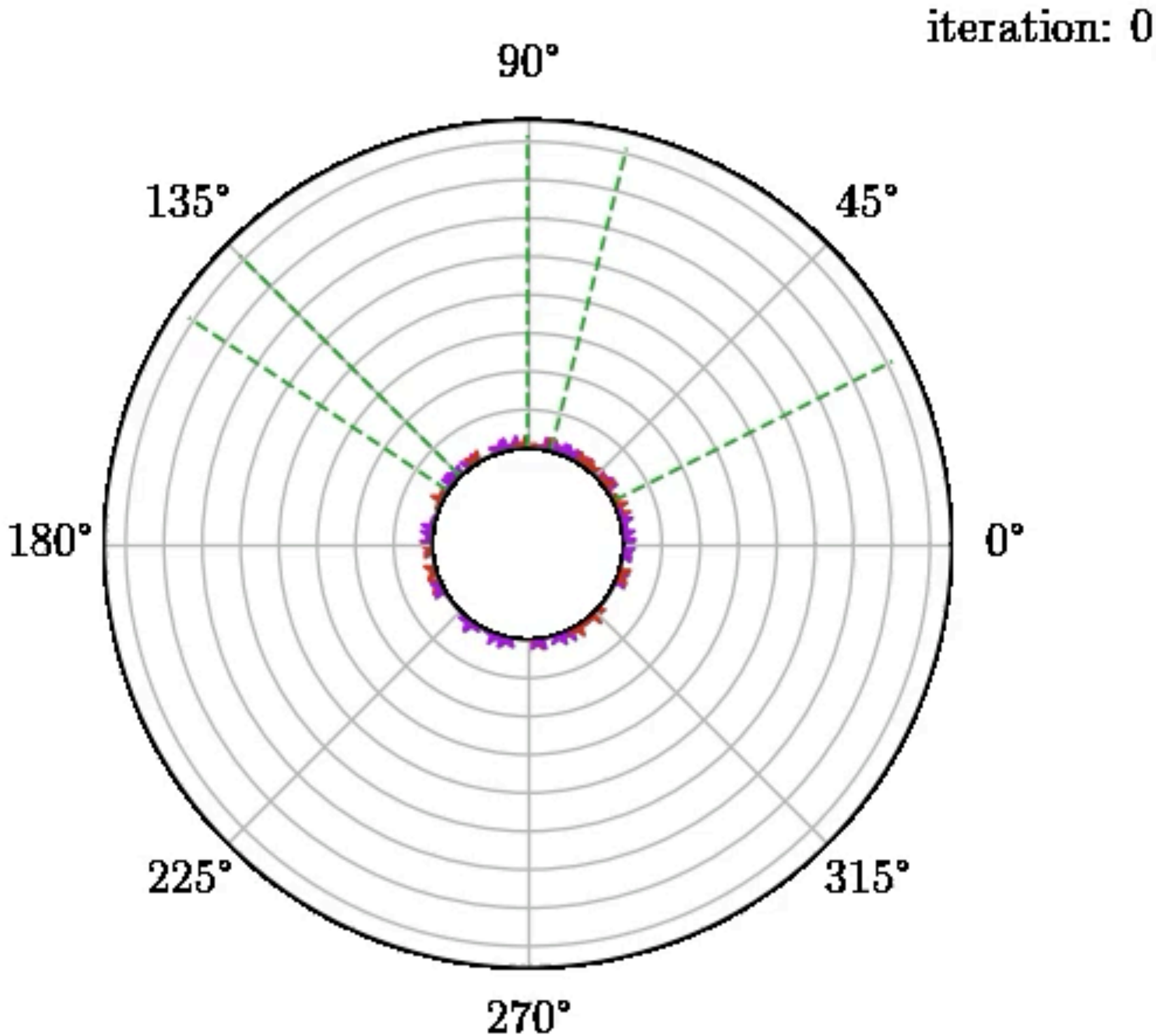
0
1
2
3
x
$-w_{j,2}/w_{j,1}$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90°
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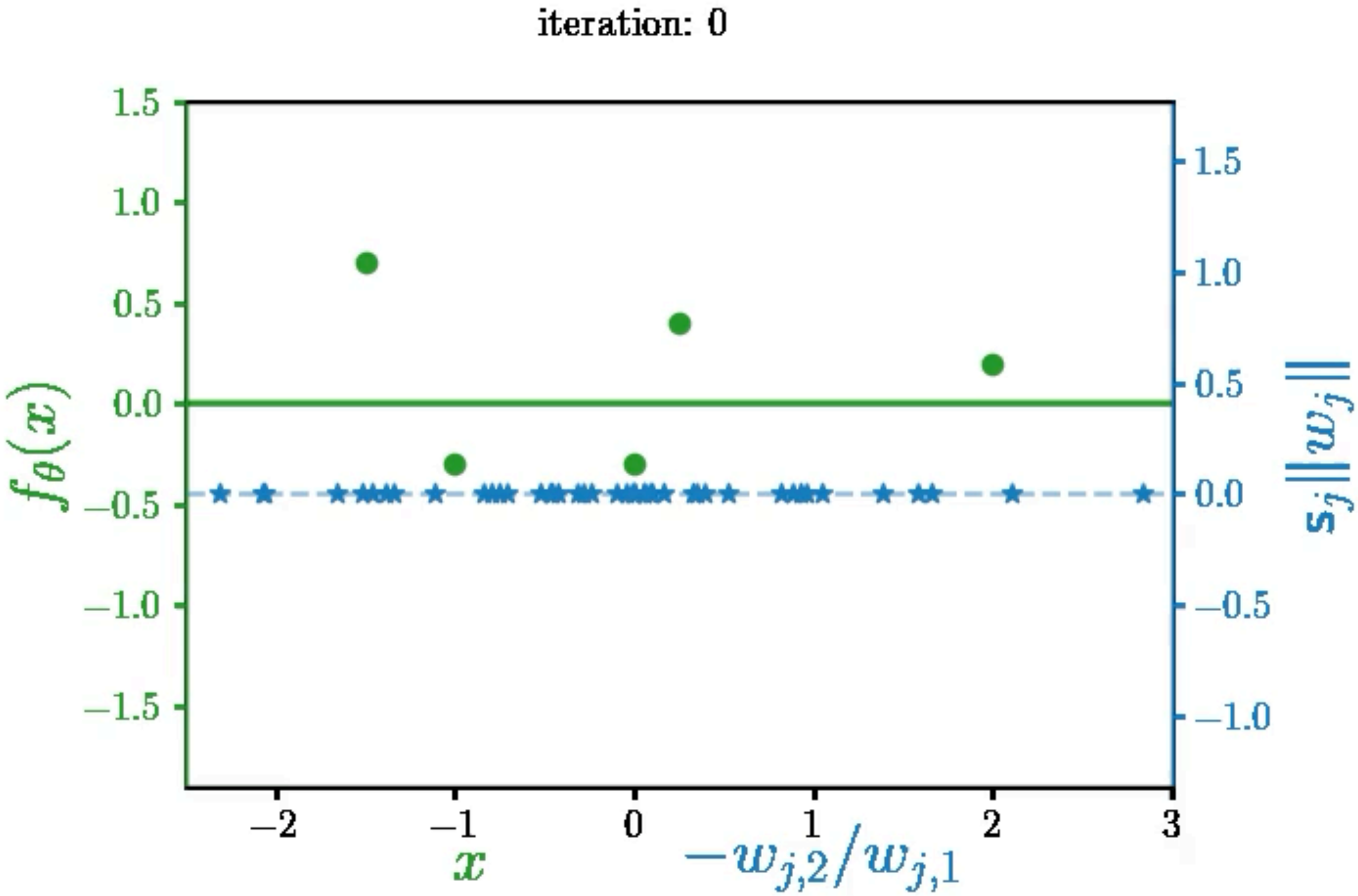
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0.5
0.0
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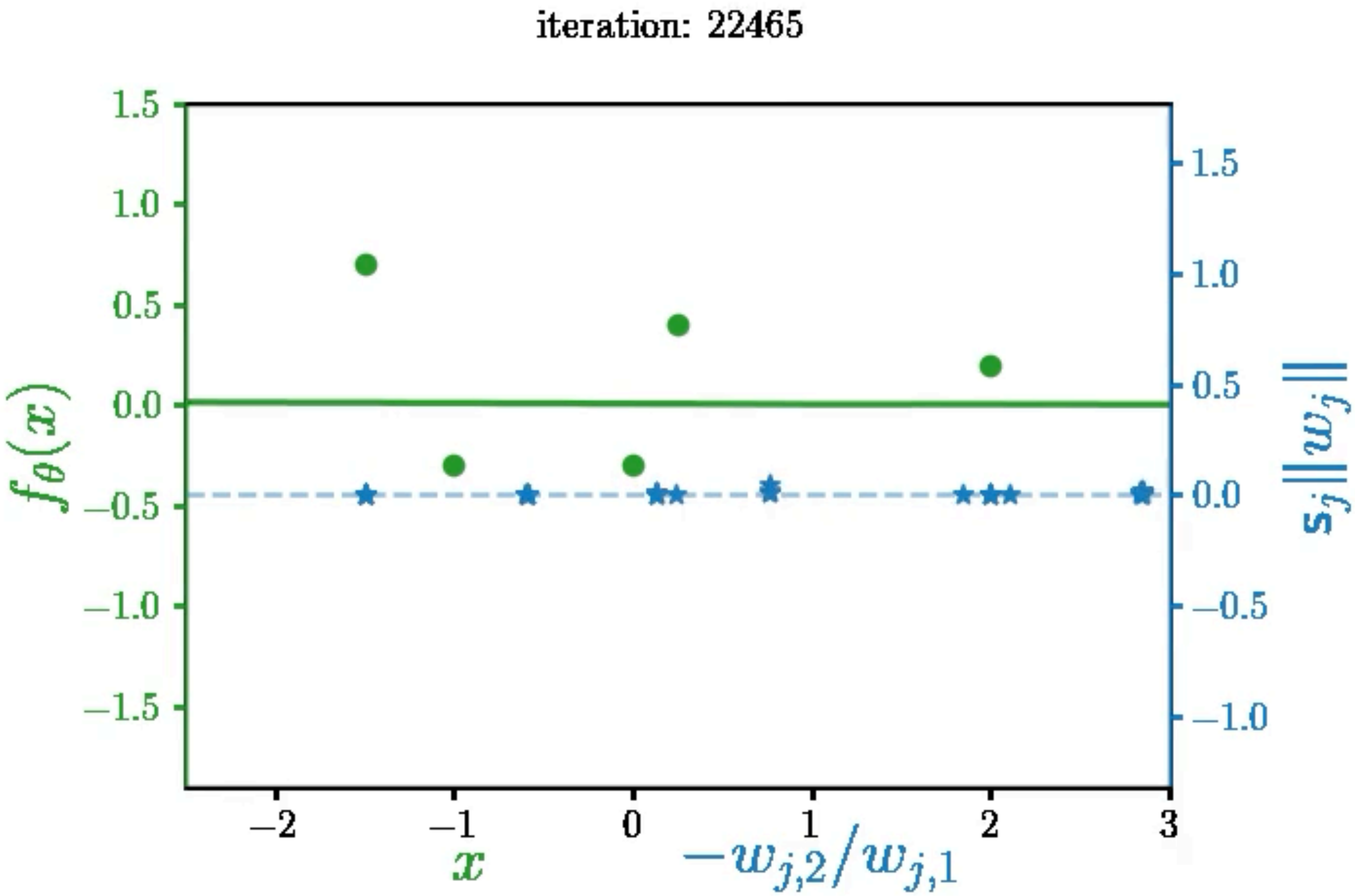
−1.5
$f_\theta(x)$
1.5
1.0
0.5
0.0
−0.5
−1.0
$s_j\|w_j\|$
−2
−1
0
1
2
3
x
$-w_{j,2}/w_{j,1}$


iteration: 22465
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0.5
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$f_\theta(x)$
1.5
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0.5
0.0
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$s_j\|w_j\|$
−2
−1
0
1
2
3
x
$-w_{j,2}/w_{j,1}$

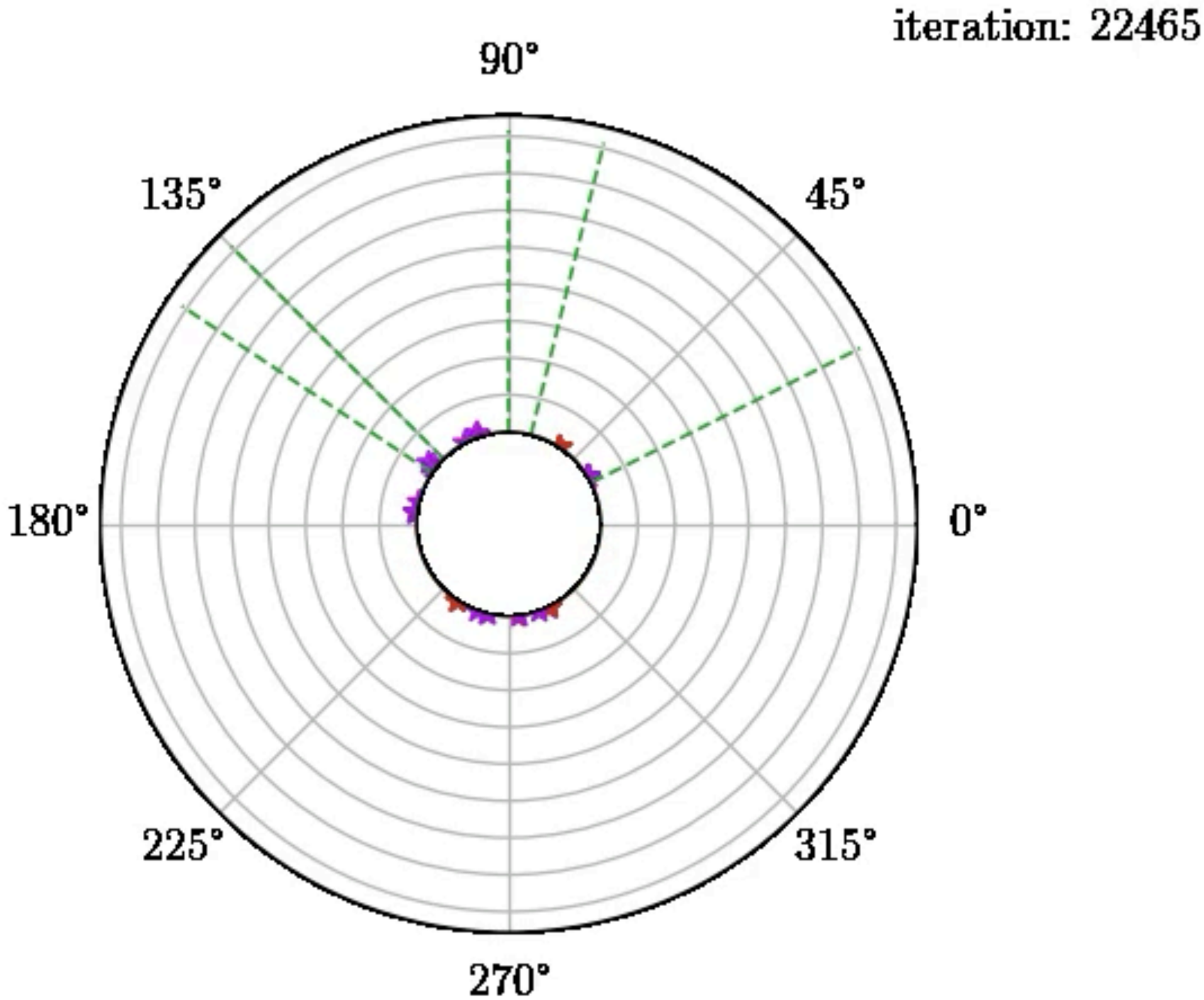
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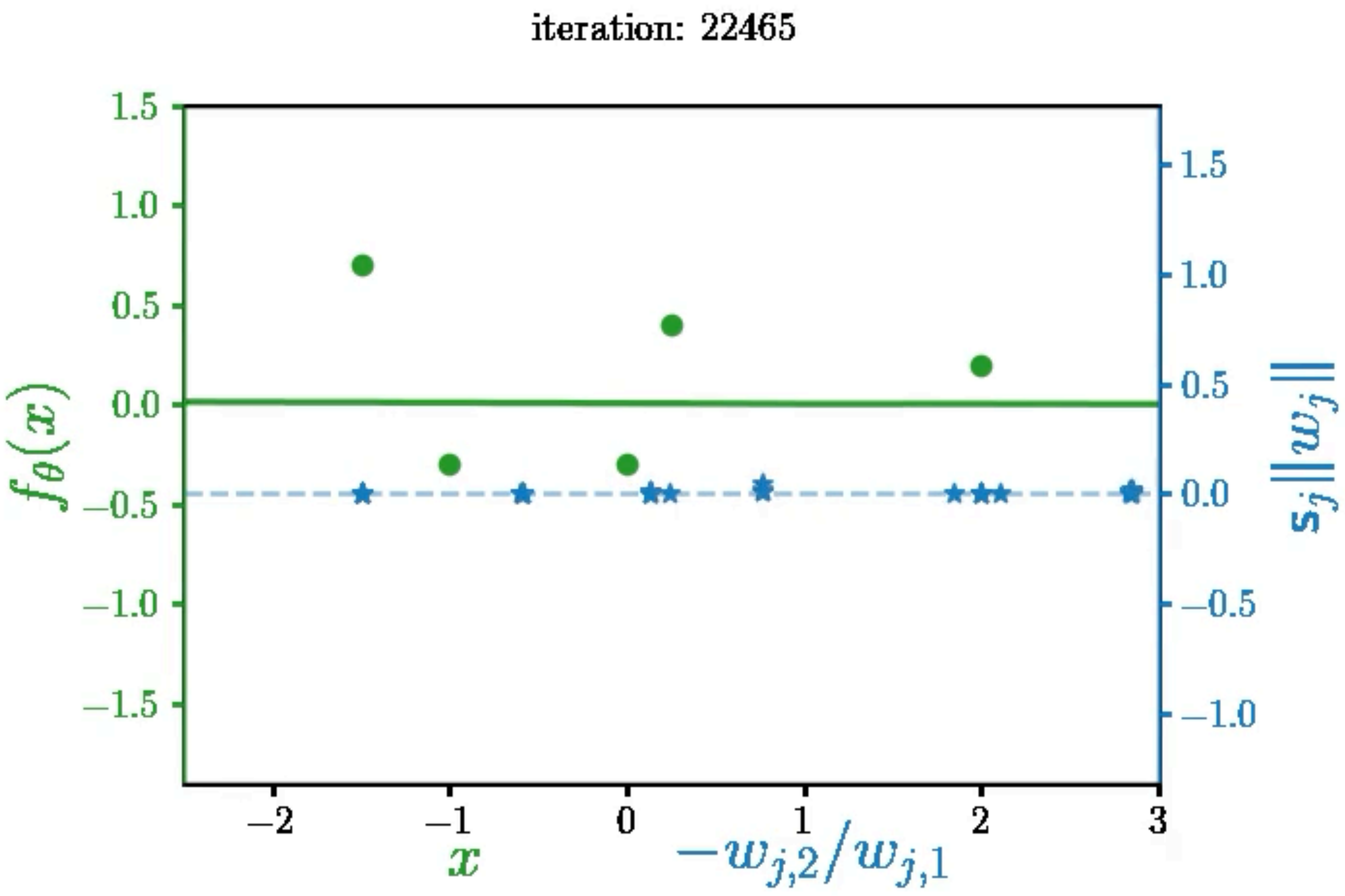
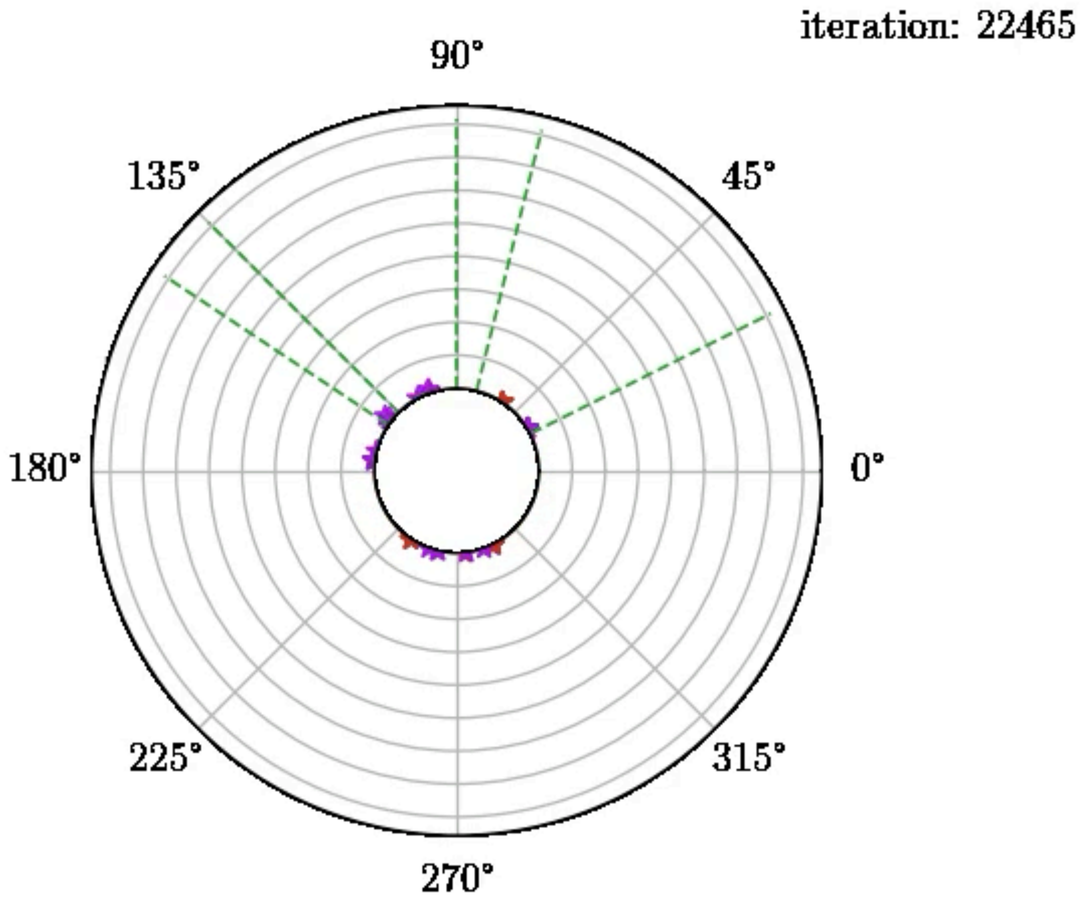
Illustration





Benefit: implicit bias

Belief

Small initialisations (rich regime) yields an implicit bias towards *simple* estimators

Simple can refer to

- Small norm parameters $||\theta||_2$
- Sparse representation of the network (e.g. small number of kinks)